

WORK №4. Optimal control design for LTI plant (Bellman's dynamic programming)

1. For linear plant

$$\dot{x} = Ax + bu, x(0)$$

and cost function

$$J = \int_0^{10} x^T(\tau)Qx(\tau) + ru^2(\tau)d\tau.$$

find the parameters of optimal controller using Bellman's dynamic programming and simulate the control system work on a given time interval.

Calculate the controller with differential Riccati equation

$$-\dot{P}(t) = A^T P(t) + P(t)A + Q - P(t)BR^{-1}B^T P(t).$$

Use the parameters for A, b, Q, r from work №3. Choose the initial conditions $x(0) = [1, 0]^T$. Plot the variables x_1, x_2, u and J .

2. Plot the cost function for changed controller parameters from the optimal ones.